

CHE 321 Chemical Reaction Engineering

Homework #1 (Due: September 5th midnight)

(please submit solutions as one combined pdf)

1. Textbook (H Scott, Fogler. 5th edition), P1-4B.

- How many pound moles of gas are in the system we have chosen for the Los Angeles basin if the temperature is 75°F and the pressure is 1 atm?
- What is the rate, $F_{CO,A}$, at which all autos emit carbon monoxide into the basin ($\frac{lb \cdot (moles \ of \ CO)}{h}$)?
- What is the volumetric flow rate ($\frac{ft^3}{h}$) of a 15 mph wind through the corridor 20 miles wide and 2000 feet high?
- At what rate, $F_{CO,S}$, does the Santa Ana wind bring carbon monoxide into the basin ($\frac{lb \cdot (moles \ of \ CO)}{h}$)?
- Assuming that the volumetric flow rates entering and leaving the basin are identical $v = v_0$ show that the unsteady mole balance on CO within the basin becomes $F_{CO,A} + F_{CO,S} - v_0 C_{CO} = V \left(\frac{dC_{CO}}{dt} \right)$
- Verify that the solution to the previous equation is $t = \frac{V}{v_0} \ln \left(\frac{[F_{CO,A} + F_{CO,S}] - v_0 C_{CO,0}}{[F_{CO,A} + F_{CO,S}] - v_0 C_{CO,f}} \right)$
- If the initial concentration of carbon monoxide in the Los Angeles basin before the Santa Ana wind starts to blow is 8 ppm ($2.04 \cdot 10^{-8} \frac{lb \ mol}{ft^3}$), calculate the time required for the carbon monoxide to reach a level of 2 ppm.
- There is heavier traffic in the L.A. basin in the mornings and in the evenings as workers go to and from work in downtown L.A. Consequently, the flow of CO into the L.A. basin might be better represented by the sine function over a 24-hour period. $F_{CO,A} = a + b \cdot \sin\left(\frac{\pi t}{6}\right)$ The time, $t = 0$, starts at midnight and $a = 35,000 \frac{lb \ mol}{h}$ and $b = 30,000 \frac{lb \ mol}{h}$. The general material balance equation now becomes $a + b \cdot \sin\left(\frac{\pi t}{6}\right) + F_{CO,S} - v_0 C_{CO} = V \left(\frac{dC_{CO}}{dt} \right)$.

HW1 Veronica Gebbia

#1 P1-4B

$$700 \text{ mi}^2 = 2 \cdot 10^{10} \text{ ft}^2$$

inversion height = 2000 ft ; 20 mi wide

$$V \text{ of air in basin} = 4 \cdot 10^{13} \text{ ft}^3$$

$$\text{conc. of CO: } 0.8 \text{ ppm or } 2.04 \cdot 10^{-10} \text{ lbmol/ft}^3$$

parts a-h:

a) pound mol of gas are in the system is $T = 75^\circ \text{F}$?

$$P = 1 \text{ atm} = 101325 \text{ Pa} ?$$

$$75^\circ \text{F} = 297.039 \text{ K}$$

$$4 \cdot 10^{13} \text{ ft}^3 \cdot \left(\frac{0.0283168 \text{ m}^3}{1 \text{ ft}^3} \right) = 1.133 \cdot 10^{12} \text{ m}^3$$

$$PV = nRT \rightarrow n = \frac{PV}{RT} \rightarrow \frac{(101325 \text{ Pa})(1.133 \cdot 10^{12} \text{ m}^3)}{(8.314 \text{ J/molK})(297.039 \text{ K})}$$

$$n = 4.65 \cdot 10^{13} \text{ mol} \left(\frac{2.2 \cdot 10^{-3} \text{ lbmol}}{1 \text{ mol}} \right) = 1.023 \cdot 10^{11} \text{ lbmol of gas}$$

b) $F_{CO,A}$ when all autos emit CO?

$$\text{total cars} = 400,000 \quad v = 3000 \text{ ft}^3/\text{h}$$

$$T = 32^\circ \text{F} = 491.67 \text{ R mol}\% = 2\% \quad R = 0.7302 \text{ ft}^3 \cdot \text{atm} / \text{lbmol} \cdot \text{R}$$

$$n = \frac{PV}{RT} \rightarrow \frac{1 \text{ atm} (3000 \text{ ft}^3/\text{h})}{(0.7302 \text{ ft}^3 \cdot \text{atm} / \text{lbmol} \cdot \text{R})(491.67 \text{ R})}$$

$$n = 8.356 \text{ lb} \cdot \text{mol} / \text{h} / \text{c} \cdot 400,000 \text{ c}$$

$$= 3342400 \text{ lb} \cdot \text{mol} / \text{h} \cdot (2/100)$$

$$= 66848 \text{ lbmol, CO} / \text{h}$$

c) what is $\dot{V}$ (ft^3/h) of a 15 mph through 20 mi wide and 2000 ft high

$$S = 15 \text{ mph} \cdot (5280 \text{ ft}/\text{h} / 1 \text{ mph}) = 79200 \text{ ft}/\text{h}$$

$$A = 2000 \text{ ft} \cdot (20 \text{ mi} \cdot (5280 \text{ ft}/1 \text{ mi})) = 2.112 \cdot 10^8 \text{ ft}^2$$

$$\dot{V} = S \cdot A \rightarrow 79200 \text{ ft}/\text{h} \cdot (2.112 \cdot 10^8 \text{ ft}^2)$$

$$\dot{V} = 1.672 \cdot 10^{13} \text{ ft}^3/\text{h}$$

d) $S = 79200 \text{ ft}/\text{h}$ $F_{\text{CO},S} = C_{\text{in}} \cdot \dot{V}$
 $A = 2.112 \cdot 10^8 \text{ ft}^2$ $C_{\text{in}} = 2.04 \cdot 10^{-10} \text{ lbmol}/\text{ft}^3$
 $\dot{V} = 1.672 \cdot 10^{13} \text{ ft}^3/\text{h}$
 $F_{\text{CO},S} = (2.04 \cdot 10^{-10} \text{ lbmol}/\text{ft}^3) (1.672 \cdot 10^{13} \text{ ft}^3/\text{h})$
 $= 3410.88 \text{ lbmol}/\text{h}$

e) $F_{\text{CO},S}$: Santa Ana wind to basin so $\dot{V} = \dot{V}_0$
 the balance: $\text{in} - \text{out} + \text{gen} - \text{cons} = \text{accum}$
 $\text{in} = F_{\text{CO},A} + F_{\text{CO},S}$ $\text{out} = \dot{V}_0 \cdot C_{\text{CO}}$
 $\text{accum} = dN_{\text{CO}}/dt = d(V C_{\text{CO}})/dt = V \cdot dC_{\text{CO}}/dt$
 it becomes:

$$F_{\text{CO},A} + F_{\text{CO},S} - \dot{V}_0 \cdot C_{\text{CO}} = V \cdot dC_{\text{CO}}/dt$$

f) verify: $t = V/\dot{V}_0 \cdot \ln(\dots)$

$$F_{\text{CO},A} + F_{\text{CO},S} - \dot{V}_0 C_{\text{CO}} = V \cdot d(C_{\text{CO}})/dt$$

$$\frac{dt}{dt} = \frac{V \cdot d(C_{\text{CO}})}{F_{\text{CO},A} + F_{\text{CO},S} - \dot{V}_0 C_{\text{CO}}}$$

$$\frac{dt}{V} = \frac{d(C_{\text{CO}})}{F_{\text{CO},A} + F_{\text{CO},S} - \dot{V}_0 C_{\text{CO}}}$$

$$dt/V = dx/(a - bx) \xrightarrow{\text{integ.}}$$

$$C_{\text{CO}} = x$$

$$\text{in} = a$$

$$\dot{V}_0 = b$$

$$\int_0^t \frac{dt}{V} = \int_{C_{\text{CO},i}}^{C_{\text{CO},f}} \frac{dx}{(a - bx)} \xrightarrow{\text{solve}} \text{sub}$$

$$\frac{\dot{V}_0 \cdot t}{V} = -\ln \left(\frac{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,f}}{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,o}} \right) +$$

combine

$$\frac{\dot{V}_0 \cdot t}{V} = \ln \left(\frac{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,f}}{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,o}} \right)$$

$$\checkmark t = V / \dot{V}_0 \cdot \ln \left(\frac{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,f}}{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,o}} \right)$$

g) $8 \text{ ppm} = 2.04 \cdot 10^{-8} \text{ lbmol/ft}^3$

find t for 2 ppm

$$V = 4 \cdot 10^{13} \text{ ft}^3 \quad \dot{V}_0 = \dot{V}_c + \dot{V}_w$$

$$F_{CO,A} = 66848 \text{ lbmol/h}$$

$$F_{CO,S} = 3410.88 \text{ lbmol/h} \quad n = 1.023 \cdot 10^{11} \text{ lbmol}$$

$$C_{CO,o} = 2.04 \cdot 10^{-8} \text{ lbmol/ft}^3$$

$$C_{CO,f} = 2 \text{ ppm} \div 1.0 \cdot 10^6 \text{ lbmol} \cdot \left(\frac{1.023 \cdot 10^{11} \text{ lbmol}}{4 \cdot 10^{13} \text{ ft}^3} \right)$$

$$= 5.115 \cdot 10^{-9} \text{ lbmol/ft}^3$$

$$\dot{V}_0 = \underbrace{3,000 \text{ ft}^3/\text{h} \cdot 400,000 \text{ c}}_{\dot{V}_c} + \underbrace{1.672 \cdot 10^{13} \text{ ft}^3/\text{h}}_{\dot{V} = \dot{V}_w}$$

$$\dot{V}_0 = 1.6721 \cdot 10^{13} \text{ ft}^3/\text{h}$$

$$t = V / \dot{V}_0 \cdot \ln \left(\frac{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,f}}{(F_{CO,A} + F_{CO,S}) - \dot{V}_0 C_{CO,o}} \right)$$

$$F_{CO,A} + F_{CO,S} = 70258.88 \text{ lbmol/h}$$

$$-\dot{V}_0 \cdot C_{CO,f} = -85527.915$$

$$-\dot{V}_0 \cdot C_{CO,o} = -341108.4$$

$$V / \dot{V}_0 = 2.392$$

$$t = 6.879 \text{ hrs}$$

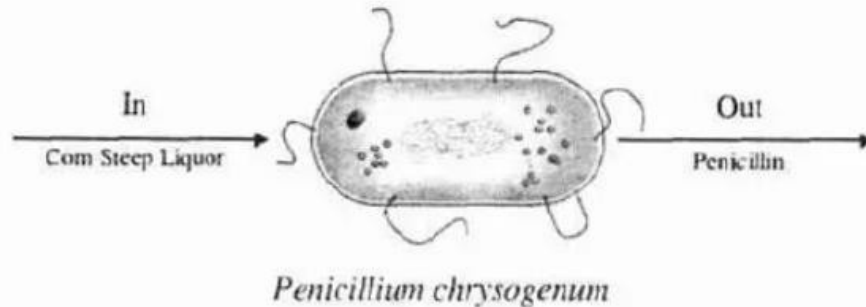
h) $F_{CO,A} = a + b(\sin(\pi t / 6))$ $a = 35,000 \text{ lbmol/h}$

$$b = 30,000 \text{ lbmol/h} \quad @ t = 0$$

$$a + b \cdot \sin(\pi t / 6) + F_{CO,S} - \dot{V}_0 C_{CO} = V(dC_{CO}/dt)$$

PolyMath?

P1-11_B We are going to consider the cell as a reactor. The nutrient corn steep liquor enters the cell of the microorganism *Penicillium chrysogenum* and is decomposed to form such products as amino acids, RNA, and DNA. Write an unsteady mass balance on (a) the corn steep liquor, (b) RNA, and (c) penicillin. Assume the cell is well mixed and that RNA remains inside the cell.



2) Unsteady Mass balance:

$$\text{in} - \text{out} + \text{gen} = \text{accum} = \frac{dN_i}{dt}$$

$$N_i = VC_i \rightarrow V \text{const} = \frac{dN_i}{dt} = V \left(\frac{dC_i}{dt} \right)$$

a) corn steep liquor (substrate (s))

$$\text{in} = FC_{s,\text{in}} \quad \text{out} = FC_s$$

$$\text{consumed} = -Vr_s$$

$$V \frac{dC_s}{dt} = FC_{s,\text{in}} - FC_s - Vr_s$$

$$\frac{dC_s}{dt} = \frac{F}{V} (C_{s,\text{in}} - C_s) - r_s$$

b) RNA (R) (inside so in/out = 0)

generation and decay only

$$\text{accum} = 0$$

$$\frac{dC_R}{dt} = G_R - D_R$$

c) Penicillin (P)

in = 0; out = a lot; Generate; decay (1st ord)

$$V \frac{dC_P}{dt} = Vr_P - FC_P - V k_{ap} C_P$$

$$\frac{dC_P}{dt} = r_P - \left(\frac{F}{V} \cdot C_P \right) - k_{ap} C_P$$

NOTE: P1-4B need to use the Smog in Los Angeles Basin Web Module.

http://umich.edu/~elements/5e/web_mod/la_basin/index.htm

Use MATLAB to solve ODE, below is the link and instructions

For MATLAB, you can access it free through UIC web store:

<https://webstore.illinois.edu/shop/product.aspx?zpid=3819> and

there is some instructions:

<https://www.mathworks.com/help/matlab/math/choose-an-ode-solver.html>

<https://www.mathworks.com/help/matlab/ref/ode45.html>